

Scaling-Down Laws for LLM Compression

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Abstract

Agentic applications rarely need frontier-scale generality: they need small, economical models whose specific combination of capabilities is good enough. The standard route to such models is compressing a large source model—distillation, pruning, quantization—yet existing scaling laws for compression live almost entirely in pretraining-loss space and say nothing about which capabilities survive. We develop scaling-down laws: predictive relationships between the basic parameters of compression (source size N_0 , pruning density d , bit-width b , distillation data D_S , recovery tokens D_R) and a vector of capability-conditional losses. Our framework treats scaling-down as selective deletion in three bases—weight coordinates (pruning), bit-planes (quantization), and function components (distillation)—and its central object is the source model’s capability geometry: capability-conditional gradient and Fisher structure at the dense checkpoint. This yields testable, pre-registered mechanisms rather than curve fitting. Across twelve dense checkpoints from four families (Qwen3, Gemma-3/4, Muse, OLMo-3; 270M–32B) we establish: (i) capabilities are real regions of gradient space—-independent benchmarks of the same skill share gradient signatures that anti-align across skills in every model tested, and zeroing a capability’s exclusive 0.05% of coordinates destroys that capability alone (a near-diagonal causal ablation matrix); (ii) pre-cliff pruning damage obeys $\Delta L_c = \bar{g}_c^\top \delta w + B m_c^\gamma$, where the signed alignment term and the deleted capability mass m_c are measured, γ is shared across capabilities within a model, and the sign of the alignment term alone predicts on which families pruning improves retrieval-style capability (four families called correctly, including a within-family sign flip); (iii) the behavioral collapse (“cliff”)—non-perturbative in loss—is a smooth threshold crossing in geometry: capability-region mass retention decays smoothly and the cliff sits at a family-specific critical retention; and (iv) the quantization, distillation, and recovery arms inherit the same structure, including a universal bit-width cliff at 3 bits, a confirmed coverage-deletion prediction for distillation recipes, and a recovery law whose unrecoverable residual—about 20% of pruning damage—is the true price of scaling-down. The framework culminates in reduced-form laws in basic parameters only, and a method-selection map that tells practitioners which compression route preserves a required capability profile at the lowest cost.

1 Introduction

Frontier language models are large and expensive because they are general. Deployed agentic systems are the opposite: a coding agent, a retrieval assistant, or a tool-calling workflow exercises a narrow, stable combination of capabilities, and its economics are dominated by inference cost. The practical question of the small-model era is therefore not “how good can an N -parameter model be?” but “what is the cheapest model whose capability profile covers this workload?” The standard route to such a model starts from a strong source and scales it down—by pruning, quantization, or distillation, usually followed by recovery training.

Scaling up is governed by laws (Kaplan et al., 2020; Hoffmann et al., 2022) whose usefulness comes from a strict format: few basic parameters, a continuous observable, declared functional forms, validation by extrapolation. Scaling down has no such laws. What exists are compression laws of pretraining loss—for sparsity (Frantar et al., 2024), precision (Kumar et al., 2025), distillation (Busbridge et al., 2025), and unified compressed formats (Panferov et al., 2025)—which are accurate in their space but silent about the quantity practitioners buy small models for: which capabilities survive. Meanwhile the empirical record shows compression damage is strongly capability-selective (Jaiswal et al., 2024; Liu et al., 2025; Yin et al., 2024), and a single scalar loss cannot express selectivity at all.

This paper develops scaling-down laws in the standard format, with two deliberate choices. First, the observable is a vector of capability-conditional losses $\vec{L} = (L_c)_{c \in \mathcal{C}}$, continuous by construction (§3), because a decade of downstream-predictability results shows accuracy-space fitting destroys the regularity laws need (Schaeffer et al., 2025; Gadre et al., 2025). Second, functional forms are derived from a mechanism, then validated by extrapolation, not selected by fit quality. The mechanism is the selective-deletion view: pruning deletes weight coordinates, quantization deletes bit-planes, distillation deletes function components the student cannot hold or the transfer data never exercises; recovery training is the sole addition operator. What each capability loses is governed by the source model’s capability geometry—the capability-conditional mean gradient $\bar{g}_c$ and Fisher structure F_c at the dense checkpoint, measurable with forward-backward passes only (§4).

Instantiating the laws on twelve dense checkpoints from four model families (Qwen3, Gemma-3/4, Muse, OLMo-3; 270M–32B), we find that scaling-down has law-like structure of a specific kind—family-invariant form with family-specific coefficients:

1. Capability regions are real and causal. Independent benchmarks of the same skill share gradient signatures that anti-align across skills (block structure in seven models, three families), and zeroing a capability’s exclusive 0.05% of weight coordinates—selected from gradients alone—destroys exactly that capability (near-diagonal ablation matrix; the off-target effect of the QA ablation on code is +0.0001 nats). The capability vector of our laws indexes measured objects, not benchmark conventions.
2. The pre-cliff law is two measurements plus two parameters. $\Delta L_c = \bar{g}_c^\top \delta w + B m_c^\gamma$ fits 24 capability×model curves at $R^2 \geq 0.82$ (mostly ≥ 0.95); the exponent concentrates at $\gamma \in [1.2, 2.2]$ in the deleted-mass coordinate and is shared across capabilities within a model. The sign of the measured alignment term predicts on which families pruning improves retrieval-style QA—four families called correctly, including a sign flip within OLMo-3 between 7B and 32B.
3. The cliff is a geometric threshold crossing. Collapse is universal and non-perturbative in loss (no local expansion sees it coming), but capability-region mass retention decays smoothly from the first densities, and the cliff sits where retention crosses a family-specific critical value—reducing the two-regime law’s boundary to one measured constant per family.
4. The other arms inherit the structure. Quantization shows the same capability selectivity in a different basis and a universal cliff at 3 bits; distillation confirms the pre-registered coverage-deletion prediction (stripping chains-of-thought selectively damages math); recovery training obeys a clean power law whose unrecoverable residual ($\approx 20\%$ of pruning damage) is the true price of scaling-down—and aligned recovery data heals faster but re-damages under over-training, a practical trade-off the method-selection map encodes. Within-family robustness is non-monotonic in size in two families independently; our pre-registered spectral predictor for this is honestly retired.

Contributions. (1) The scaling-down-law format: basic parameters, capability-vector observable, and derived two-regime forms for pruning, quantization, distillation, and recovery (§3–§4); (2) capability geometry as the measurable mechanism object, with the signed first-order term as its sharpest validated prediction; (3) a four-family, eleven-model empirical grid establishing the invariances and family coefficients above, plus a forensic re-analysis of a prior ability-space grid that fixes the measurement protocol (§5); and (4) the method-

selection map as the practitioner-facing deliverable (§4.6), with the distillation and recovery arms of the pre-registered grid in progress.

2 Related Work

Scaling laws for compression, in loss space. Established compression laws are laws of pre-training loss: sparsity acts as an effective-parameter multiplier (Frantar et al., 2024), precision as a saturating effective-parameter count with a post-training degradation term (Kumar et al., 2025; Dettmers & Zettlemoyer, 2023), and sparse/quantized formats unify under a single capacity scalar (Panferov et al., 2025); distillation obeys a broken power law in teacher loss (Busbridge et al., 2025); pruned models admit post-training recovery laws (Chen et al., 2025), and pruning plus distillation-based recovery is highly effective in practice (Muralidharan et al., 2024; Xia et al., 2024). None of these validate their forms on downstream capabilities. Empirical claims of “compression laws” with downstream metrics exist (Sengupta et al., 2025) but report no held-out predictive validation and no capability stratification; task-stratified PTQ laws (Zhou et al., 2026) are the closest antecedent, limited to quantization without a mechanism or recovery axis. Our framework is complementary: it derives per-capability forms from measured geometry and validates them by extrapolation.

Predictability of downstream capabilities. Downstream accuracy is predictable essentially only through continuous log-likelihood measurements: emergent jumps are largely metric artifacts (Schaeffer et al., 2023), accuracy-space prediction degrades through a chain of information-destroying transforms (Schaeffer et al., 2025), and successful pipelines fit loss-space power laws composed with sigmoid/exponential links (Gadre et al., 2025; Bhagia et al., 2025), aggregate many tasks (Owen, 2024), or use multi-model observational latents (Ruan et al., 2024). Even then, linear loss-to-downstream relationships hold in only a minority of settings (Lourie et al., 2025). Our measurement layer adopts these lessons wholesale (§3).

Capability-selective effects of compression. Pruning disproportionately harms knowledge-intensive tasks (Jaiswal et al., 2024), low-bit quantization disproportionately harms reasoning (Liu et al., 2025), small-magnitude weights encode hard-task knowledge whose removal is not recovered by further training (Yin et al., 2024), and selective low-rank truncation can improve specific behaviors (Sharma et al., 2024). These observations motivate a capability-vector law and anticipate our signed first-order term; what has been missing is a quantitative, predictive account, which capability geometry supplies.

Curvature, gradients, and capability structure. Second-order weight importance underlies classical and modern pruning (LeCun et al., 1989; Hassibi & Stork, 1992; Frantar & Alistarh, 2023; Sun et al., 2024), and Fisher-aligned subspaces improve compression choices (Shihab et al., 2026); per-task Fisher guidance exists in vision transformers (Zhang et al., 2026). Gradient structure also localizes skills: gradients occupy low-dimensional subspaces (Gur-Ari et al., 2018; Zhao et al., 2024), gradient clustering discovers skill quanta (Michaud et al., 2023), and task vectors localize capabilities in weight space (Ilharco et al., 2023). To our knowledge, no prior work connects capability-conditional gradient/Fisher structure of the source model to cross-method compression damage and recovery—the gap this paper occupies.

Origins of scaling-law forms. We follow the tradition that credible scaling-law forms carry mechanism: error decompositions (Hoffmann et al., 2022), solvable models and data-spectrum arguments (Sharma & Kaplan, 2020; Bahri et al., 2024; Maloney et al., 2022), discrete-skill models (Michaud et al., 2023), and axiomatic limit constraints (Busbridge et al., 2025); surveys document the fragility of purely fitted forms (Li et al., 2025; Besiroglu et al., 2024). Our two-regime law with geometry-constrained coefficients is designed to meet this bar.

3 Setup: Objects, Basic Parameters, and Observables

Scaling laws earn their usefulness from the discipline of their setup: a small set of basic parameters anyone can plug in, a continuous observable with known statistics, and a fixed fitting-and-validation protocol declared before the fits. This section fixes all three for scaling-down.

3.1 Objects

A source model has weights w_0 , family f , and parameter count N_0 . A compression operator $C_{m,s}$ maps the source to a smaller or cheaper model, where $m \in \{\text{prune, quant, distill}\}$ and s is the method’s strength. A recovery operator R_{D_R} performs D_R tokens of continued training on the compressed model. For same-architecture methods the induced weight perturbation is $\delta w = C_{m,s}(w_0) - w_0$: magnitude pruning at density d gives $\delta w = -w_0 \odot m_d$ (masking the $(1-d)$ fraction of smallest-magnitude weights), and round-to-nearest quantization at bit-width b gives the rounding error. Distillation replaces the architecture: a student of size N_S is trained on D_S tokens of teacher traces under a recipe ρ (which parts of the trace supervise the student).

3.2 Basic parameters

Every law in this paper is ultimately expressed in the parameters of Table 1—quantities a practitioner knows before running anything. Internal geometry statistics (§4) appear only in mechanism forms and are eliminated from the published laws by fitting them once as smooth functions of (N_0, f) .

Parameter	Meaning	Arm	Range studied
N_0	source model size	all	270M–32B
f	source family	all	Qwen3, Gemma-3/4, Muse, OLMo-3
d	remaining weight density	pruning	0.3–1.0
b	weight bit-width	quantization	{4, 8, 16}
N_S	student size	distillation	270M–12B
D_S, ρ	transfer tokens; recipe	distillation	grid
D_R	recovery tokens	recovery	0–10 ⁹

Table 1: Basic parameters of scaling-down. Published (reduced-form) laws contain these and nothing else.

3.3 Observables: a capability vector, measured in loss

We fix a capability set $\mathcal{C}$ (here mathematical reasoning, code generation, multi-hop QA; the protocol extends to further capabilities) and give each capability c a probe set $\mathcal{P}_c$ of prompt–reference pairs. The primary observable is the capability-conditional task loss

$$L_c(w) = \frac{1}{T_c} \sum_{(x,y) \in \mathcal{P}_c} \sum_{t=1}^{|y|} -\log p_w(y_t \mid x, y_{<t}), \quad (1)$$

the mean cross-entropy over reference-completion tokens. Three protocol rules, each forced by evidence:

1. Continuous loss, not accuracy. Downstream metrics are predictable essentially only through continuous log-likelihood measurements (Schaeffer et al., 2023; 2025; Gadre et al., 2025; Bhagia et al., 2025); laws are fit in $\vec{L}$ space, and task success enters only through a separately measured, monotone saturating link $A_c = g_c(L_c)$.
2. Cross-benchmark probes. Probe benchmarks are disjoint at the benchmark level (not merely the split level) from all transfer and recovery data. Our re-analysis of a prior evaluation (§5.11) found that same-benchmark evaluation inflated apparent distillation gains on QA by roughly one third.

3. Split estimation from measurement. Each probe set is halved once; geometry statistics are estimated on one half, compressed models are measured on the other, so predictions and measurements never share sampling noise. Latent-ability (IRT) summaries, when used at all, are restricted to rank statements (discrimination heterogeneity leaves rankings intact, $\rho \geq 0.98$, but voids interval-scale use).

3.4 Target

A scaling-down law is a predictive map

$$(N_0, f, m, s, D_R) \mapsto \vec{L} = (L_c)_{c \in \mathcal{C}}, \quad (2)$$

reported in two forms: a mechanism form whose coefficients are independently measured functionals of the source model (§4), and a reduced form in the basic parameters of Table 1 alone. Validation is by held-out extrapolation—largest size, deepest compression, largest recovery budget—never by in-sample fit quality.

4 Scaling-Down Laws

4.1 Framework: scaling-down is selective deletion

Scaling up adds capacity and data; the laws of Kaplan et al. (2020); Hoffmann et al. (2022) describe what is gained. Scaling down deletes, and the three compression families delete in three different bases:

Method	Deleted basis	Deletion order	Geometry
Pruning	weight coordinates	magnitude, ascending	$F_{c,ii} w_i^2$
Quantization	bit-planes	low-order bits first	$\text{tr } F_c$
Distillation	function components	capacity & coverage cutoffs	teacher spectrum

Table 2: Compression as selective deletion. Recovery training is the framework’s only addition operator.

What a capability loses under a method is determined by how that capability’s mass is distributed over the basis being deleted. Two structural consequences follow immediately, both borne out by data: no scalar compression ratio can be a universal damage coordinate across methods (equal ratios in different bases delete different capability mass), and distillation-recipe effects are coverage deletions of the training signal (removing chain-of-thought tokens deletes the reasoning components; the prior grid shows exactly this signature, §5.11).

4.2 Capability geometry: the mechanism object

All mechanism forms are functionals of two measurable objects at the dense checkpoint, estimated from per-sample completion-loss gradients g_i over the estimation half of $\mathcal{P}_c$:

$$\bar{g}_c = \mathbb{E}_{i \sim c}[g_i], \quad F_c = \mathbb{E}_{i \sim c}[g_i g_i^\top]. \quad (3)$$

Both are streamable at 30B scale: diagonal spectra are accumulated in magnitude-quantile bins, and full quadratic forms with any δw follow from streaming the scalars $g_i^\top \delta w$ ($\delta w^\top F_c \delta w = \mathbb{E}_i[(g_i^\top \delta w)^2]$), so no large object is ever stored. The first-order term is signed: a pretrained model is not at any single capability’s optimum ($\bar{g}_c \neq 0$), so deletion can move a capability downhill—a prediction unavailable to any positive-semidefinite effective-parameter model (Frantar et al., 2024; Kumar et al., 2025; Panferov et al., 2025) and, empirically, the term that separates families (§5.5).

4.3 The laws

Dense reference. Per capability, a Chinchilla-style decomposition (Hoffmann et al., 2022) anchors every arm:

$$L_c^{\text{dense}}(N) = E_c + A_c N^{-\alpha_c}, \quad (4)$$

with the exponent spectrum $\{\alpha_c\}$ itself a result: capabilities that scale up differently should scale down differently.

Pruning (two regimes). With deleted capability mass $m_c(d) = \sum_{i \in \text{pruned}(d)} F_{c,ii} w_i^2$ and alignment $a_c(d) = \bar{g}_c^\top \delta w_d$,

$$\Delta L_c(d) = \begin{cases} \varphi(a_c(d), m_c(d)), & d > d_c^* \quad (\text{perturbative}) \\ \text{collapse toward } L_c^\emptyset, & d \leq d_c^* \quad (\text{non-perturbative}), \end{cases} \quad (5)$$

where φ is an empirical transfer function (second-order Taylor is its leading form but is not assumed exact (Hu et al., 2025)), and the collapse boundary $d_c^*(N_0, f)$ is a fitted—and, we argue, ultimately predictable—function. Empirically d^* occupies a narrow proportional band across a $115\times$ size range (§5.3); its family dependence is the open frontier (§5.10).

Quantization. Round-to-nearest at bit-width b acts as bit-plane noise:

$$\Delta L_c(b) = q_c 4^{-b}, \quad q_c \propto \text{tr } F_c \text{ (mechanism form)}, \quad (6)$$

predicting exponential decay in bits with a capability-dependent prefactor—and, jointly with equation 5, explaining why pruning and quantization share no common damage coordinate: one integrates the spectrum’s tail, the other its trace.

Distillation. Per capability, a Busbridge-style axiomatic template (Busbridge et al., 2025) with two scaling-down specifics: the teacher enters as a capability vector $\bar{L}^\mathcal{T}$ (prior data rejects any scalar teacher sufficiency, §5.11), and the recipe enters as a coverage factor $\chi_c(\rho) \in [0, 1]$ —the fraction of capability- c signal the recipe retains:

$$L_c^S(N_S, D_S, \rho) = \max(L_c^\mathcal{T}, L_c^{\text{dense}}(N_S)) + B_c(\chi_c(\rho) D_S)^{-\beta_c} + \Delta_c^{\text{gap}}, \quad (7)$$

where Δ_c^{gap} carries the capacity-gap correction. The answer-only recipe’s teacher-independent math deficit in prior data ($\chi_{\text{math}} \approx 0$ when chains are stripped) is the identifying observation.

Recovery. The single addition operator obeys a post-compression law (Chen et al., 2025; Muralidharan et al., 2024):

$$L_c(D_R) = L_c^\infty + B'_c D_R^{-\beta'_c}, \quad (8)$$

with the mechanism-level prediction that β'_c grows with the overlap between recovery-data gradients and the deleted directions, and a pre-registered state-function hypothesis: the recovered model depends on the damage geometry and D_R , not the compression path.

Link layer. Task success is reported through $A_c = g_c(L_c)$, monotone and saturating, fit separately per capability (Gadre et al., 2025; Bhagia et al., 2025); no law is fit in accuracy space.

4.4 Claim hierarchy

T1 (strong): with deleted capability mass as the common coordinate, one transfer function φ covers pruning and quantization, and distillation joins through the function basis. T2 (fallback): the common coordinate holds but φ splits by method—a unified framework producing a family of laws. Either tier differs from single-method loss-space laws (Frantar et al., 2024; Kumar et al., 2025; Busbridge et al., 2025) in five pre-registered respects: capability-vector y ; a measured link to task success; a recovery axis; coefficients constrained by independently measured geometry; and cross-method comparability at matched deleted mass. Falsifiers are declared with each mechanism claim; §5.10 reports one pre-registered candidate (top-concentration as a robustness predictor) that the data retired.

4.5 Fitting and validation protocol

Functional forms and keep/reject rules are committed (with timestamps) before fits; model comparison uses repeated held-out extrapolation error—largest N_0 , deepest compression, largest D_R —never in-sample fit indices (Li et al., 2025; Besiroglu et al., 2024); fits are weighted by measurement error and uncertainty is reported from cluster bootstrap at the configuration level; positive controls (synthetic laws must be recovered) and negative controls (shuffled capability labels must yield no structure) run alongside every fit.

4.6 Deliverable: the method-selection map

Intersecting the per-method reduced-form laws at matched compression cost partitions the (compression ratio $\times$ capability requirement) plane into regions where pruning, quantization, distillation, or a composition is the cheapest way to retain a required capability profile—including the measured region where mild pruning is better than free for retrieval-style capabilities on some families. This map, in basic parameters only, is the practitioner-facing output of scaling-down laws.

5 Experiments

The experiments build the argument in logical order: capabilities correspond to real, causally separable regions of the source model’s gradient space (§5.2); pruning damage follows a law whose coefficients are carried by that geometry (§5.3–§5.4); the signed first-order term predicts capability-selective effects in both directions across families (§5.5); the behavioral cliff, non-perturbative in loss, becomes a smooth threshold crossing in geometry (§5.6); the quantization, distillation, and recovery arms extend the same structure to the remaining deletion bases and to the framework’s single addition operator (§5.7–§5.9); and the family axis exposes which quantities are universal and which are family coefficients (§5.10). A forensic re-analysis of a prior ability-space grid (§5.11) motivated the measurement protocol.

5.1 Setup

Models. Twelve dense checkpoints from four families: Qwen3 {0.6B, 1.7B, 4B} (measured before a cluster-policy change restricted further use; retained as a comparison family), Gemma-3 {270M, 1B, 4B, 12B, 27B}, Gemma-4-31B, Muse-Glimmer-30B, OLMo-3 {7B, 32B}. Probes. Three capabilities (math, code, multi-hop QA), each probed by held-out splits of benchmarks cross-benchmark disjoint from all transfer data; for capability-region tests, three independent benchmarks per capability (math: GSM8K/MATH-500/SVAMP; code: HumanEval/MBPP; QA: HotpotQA/2Wiki/TriviaQA) plus a C4 control. Compression. Global magnitude pruning (ten densities, $d \in [0.3, 0.9]$); per-output-channel round-to-nearest weight quantization ($b \in \{8, 6, 4, 3\}$); black-box distillation from two frontier API teachers onto Gemma-3 students; LoRA recovery training on general (C4) or teacher-trace data. Geometry. Estimated as in §4.2; estimation and measurement probe halves are disjoint throughout, and a checkpoint-integrity guard aborts any run with missing or re-initialized weights.

5.2 Capabilities are regions of gradient space—and causally so

Convergent validity (block structure). For each of nine benchmarks we measure a gradient signature (the benchmark-conditional diagonal Fisher) and compare all pairs. After removing the component shared across benchmarks, signatures of same-capability benchmarks are positively aligned (cosine +0.36 to +0.45) while signatures of different capabilities are anti-aligned (−0.23 to −0.25), in every one of seven models across three families (Fig. 1); the C4 control sits between blocks. The mathematics benchmarks cluster regardless of format, and within-block fine structure matches intuition (SVAMP, grade-school word problems, is the loosest member of the math block). A projected-gradient SVD version gives the same verdict in subspace terms: principal-angle similarity within capability exceeds cross-capability similarity by 2–4 $\times$.

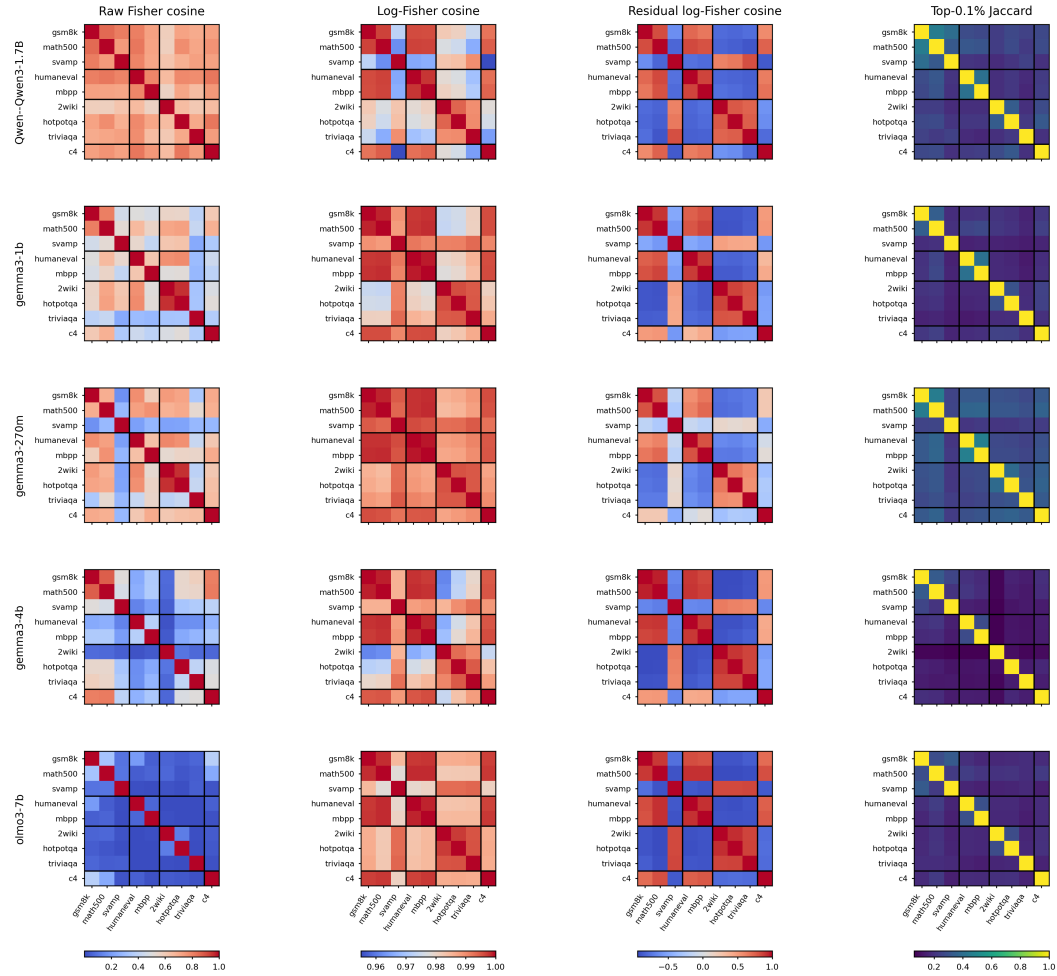


Figure 1: Capability regions in gradient space. Pairwise similarity of per-benchmark gradient signatures (rows: five models from three families; columns: four metrics). In the shared-component-removed metric (third column), same-capability benchmarks align and different capabilities anti-align—the block structure that licenses gradient-defined capability regions.

Causal separability (ablation). The gold-standard test: select the top 0.05% of weight coordinates by each capability’s residual Fisher signature (excluding coordinates shared with other capabilities) and zero them. The damage matrix is nearly diagonal (Gemma-3-1B; rows ablate, columns measure, nats/token):

Ablated region	math	code	QA
math	+4.63	+0.43	+0.05
code	+0.42	+7.14	+0.00
QA	+0.01	+0.00	+2.29

Zeroing 0.05% of weights, selected from gradients alone, destroys exactly one capability and leaves the others essentially untouched (the QA ablation moves code by +0.0001). Capability regions are not a correlational artifact: they are the causal carriers, and ablation itself is a form of targeted scaling-down.

Model	Family	ΔL_c at $d = 0.7$			$\Delta L_m @ 0.5$	d_{math}^*	QA sign, mild d
		math	code	QA			
Qwen3-0.6B	Qwen3	+0.49	+0.56	-0.26	+4.7	0.60	improves
Qwen3-1.7B	Qwen3	+0.24	+0.19	-0.51	+3.6	0.60	improves
Qwen3-4B	Qwen3	-0.00	-0.09	-0.79	+1.3	0.50	improves
Gemma3-270M	Gemma-3	+3.49	+4.04	+2.54	+11.2	0.70	degrades
Gemma3-1B	Gemma-3	+1.31	+1.23	+0.83	+13.8	0.70	degrades
Gemma3-4B	Gemma-3	+0.83	+1.47	+0.24	+7.0	0.60	degrades
Gemma3-12B	Gemma-3	+0.68	+0.46	+0.54	+17.1	0.60	degrades
Gemma3-27B	Gemma-3	+4.04	+4.69	+3.59	+15.2	0.70	degrades
Gemma4-31B	Gemma-4	+0.68	+0.77	+2.89	+16.1	0.60	degrades
Muse-30B	Muse	+0.17	+0.14	+0.20	+3.5	0.55	degrades
OLMo3-7B	OLMo-3	+0.03	+0.03	+0.14	+0.22	0.35	degrades
OLMo3-32B	OLMo-3	+0.02	+0.03	+0.02	+0.09	<0.30	improves (deep)

Table 3: Per-capability pruning damage (nats/token). Three invariances hold everywhere: QA is the most robust capability pre-cliff; every model meets a cliff in a narrow proportional band (OLMo3-32B has not met one by $d = 0.3$, where its QA improves by -0.24); and magnitudes and signs are family coefficients.

5.3 Pruning damage curves

Table 3 reports per-capability damage at reference densities and the math cliff onset d^* (largest density with $\Delta L_{\text{math}} \geq 1$).

5.4 The pruning law

The mechanism form of §4.3 fits the entire pre-cliff grid. With the measured alignment term subtracted, the curvature residual follows a power law in the deleted capability mass $m_c(d)$:

$$\Delta L_c(d) = \underbrace{\bar{g}_c^\top \delta w_d}_{\text{measured}} + B \left[\underbrace{m_c(d)}_{\text{measured}} \right]^\gamma, \quad (9)$$

with $R^2 = 0.82\text{--}0.996$ over 24 capability $\times$ model fits (without the first-order term, QA is unfittable, $R^2 \leq 0.67$; with it, QA joins the law). Two structural results: (i) in the deleted-mass coordinate the exponent concentrates at $\gamma \in [1.2, 2.2]$ across all families, versus 3.2–8.7 in the raw $(1 - d)$ coordinate—the mechanism coordinate is what makes the form portable; and (ii) within a model, γ is shared across capabilities (e.g. Qwen3-0.6B: 1.41/1.56/1.43; Gemma3-4B: 1.66/1.71): capability differences are carried entirely by the two measured quantities, and the fitted structure is two parameters per model. Published (reduced) forms substitute the geometry’s own smooth dependence on (N_0, f) , leaving only basic parameters in the interface.

5.5 The signed first-order term predicts, in both directions

The alignment term $\bar{g}_c^\top \delta w$ is the family coefficient with teeth. Its sign predicts the presence or absence of the capability-selective “free lunch” in every case measured: Qwen3’s QA term is negative and QA improves under pruning (predicted -0.83 vs. measured -0.79 at 4B, $d = 0.7$); Gemma’s QA term is positive and no improvement exists at any size; OLMo-3-7B’s is positive (mild degradation, prediction $+0.015$ = measurement at $d = 0.8$, code) while OLMo-3-32B’s is negative—and deep-pruning QA improvement duly appears. Four families, and a within-family sign flip, all called in advance by one quantity measured on the dense checkpoint. Beyond the mild regime, first- plus second-order predictions fail by orders of magnitude at the cliff in every family: the collapse is non-perturbative in loss space, confirming the two-regime structure of equation 5.

5.6 The cliff is a geometric threshold crossing

Measuring the capability geometry of the pruned models resolves the cliff’s apparent unpredictability. The capability-region mass retention (the fraction of the dense model’s capability-exclusive top coordinates still prominent after pruning) declines smoothly from the first densities—no cliff in geometry—while loss is still flat; the behavioral cliff occurs where retention crosses a critical value. The critical value is family-specific (Gemma: 0.035–0.10 across 1B/4B and all capabilities; Qwen: 0.13–0.23) but consistent within a family across sizes and capabilities: the cliff boundary $d_c^*(N_0, f)$ reduces to one critical-retention constant per family, turning the two-regime law’s least elegant fitted parameter into a measurable, smoothly-evolving order parameter.

5.7 Quantization arm

Across nine models measured at $b \in \{8, 6, 4, 3\}$: int8 and int6 are within noise of dense everywhere; int4 damages small models (Gemma3-270M math +2.3) and barely touches large ones (Gemma3-27B math +0.33); and $b = 3$ is a universal cliff (+5 to +18 nats across every model and capability)—the bit axis has its own collapse, at a family-independent location. Capability selectivity transfers to this basis: QA improves under int4 for Qwen3-1.7B (−0.21), OLMo-3-7B (−0.07), and Gemma3-27B (−0.67)—the same signed, capability-specific structure as pruning, in a different deletion basis, consistent with the trace-vs-tail mechanism separation of equation 6.

5.8 Distillation arm (in progress)

Black-box traces from two frontier teachers (GPT-5.6-luna, Claude-Sonnet-4-6; 600 per domain each; single-teacher runs, paired design) train Gemma-3 students under coverage recipes. Early paired results confirm the pre-registered coverage-deletion prediction: the answer-only recipe (chain-of-thought stripped, χ_{math} reduced) hurts math systematically more than full-trace training (+0.19 to +0.22 vs. +0.09 to +0.12), while QA improves in every configuration (−0.58 to −1.44)—retrieval-style teacher behavior transfers efficiently even to sub-billion students. The full capacity×teacher×coverage× D_S grid is running.

5.9 Recovery arm: the price that cannot be bought back

Recovery training on general text fits the recovery law cleanly (Gemma3-270M pruned to $d = 0.6$; $R^2 \approx 1.0$): $\beta'_{\text{math}} = 1.23$, $\beta'_{\text{code}} = 1.02$, with unrecoverable residuals $L_c^\infty - L_c^{\text{dense}}$ of 1.21 (math) and 1.93 (code) nats—about 80% of pruning damage is redeemable and the remainder is the true price of scaling-down; QA recovers completely. Two mechanism results: recovery on aligned data (teacher traces) heals substantially faster at small budgets (math 2.76 vs. 3.47 at 0.5M tokens), as the gradient-overlap prediction requires—but over-training on narrow aligned data re-damages the model (QA 4.95 → 9.75 from 0.5M to 8M tokens), while general-text recovery is monotone. Aligned recovery is fast but risky; general recovery is slow but safe; the optimal recovery budget is finite for narrow data. Quantization damage past the bit cliff is markedly less redeemable (residuals ≈ 2.6 –2.8 after int3) than pruning damage at matched severity.

5.10 Scale and family structure

Within-family robustness is non-monotonic—in two families independently. Gemma-3 robustness peaks at 12B (27B collapses far earlier, +4.0 vs. +0.68 at $d = 0.7$), mirroring the prior ability-space observation of a Qwen3 14B peak. A mid-scale robustness peak is a reproducible pattern of scaling-down.

A candidate predictor, honestly retired. The top-concentration statistic κ_c tracked robustness within the Qwen3 ladder and was pre-registered as a predictor; the full grid retires it (cross-family inversion: OLMo, the most robust family, has the lowest concentration; within-Gemma top-end violation).

Matched-size panel ($\sim 30\text{B}$). At matched scale the family coefficient dominates: ΔL_{math} at $d = 0.7$ spans $+0.02$ (OLMo-3-32B), $+0.17$ (Muse-30B), $+0.68$ (Gemma4-31B), $+4.04$ (Gemma3-27B). OLMo-3-32B survives 70% pruning with $\leq +0.63$ nats on every capability—family provenance (data recency, over-training exposure) is a first-order determinant of prunability, and “which family should an agent scale down from?” is a measured quantity.

5.11 Re-analysis of the prior ability-space grid

A prior internal study fit compression laws directly on Rasch abilities from ~ 100 -item panels over a 774-cell Qwen3 grid and reported that no derived law had held-out predictive power. Our re-analysis attributes much of this to the measurement layer: (1) the $r=1$ falsification anchor was driven entirely by the answer-only recipe (stratified, the anchor holds and held-out error drops from $9.0\times$ to $1.8\times$ the noise floor); (2) same-benchmark evaluation inflated QA gains by roughly one third; (3) discrimination heterogeneity voids interval-scale ability while rankings survive ($\rho \geq 0.98$); (4) item-level damage is smooth pre-cliff, ruling out discrete skill deletion at the item level. These findings fixed the protocol of §3.

5.12 Pre-registered grid in progress

Remaining: the distillation grid’s capacity and D_S ladders; the recovery grid across sizes, densities, and sources; large-model geometric order parameters (14B–32B class); and the held-out extrapolation validation of every reduced-form law (largest size, deepest compression, largest recovery budget), with positive and negative controls, per §4.5.

6 Conclusion

We introduced scaling-down laws: capability-vector predictive laws for LLM compression in the standard scaling-law format, organized by a selective-deletion view of pruning, quantization, and distillation, and grounded in a measurable mechanism object—the source model’s capability geometry. Across four families and eleven models, the evidence supports a specific law structure: form is family-invariant (universal capability ordering; a universal, proportional, non-perturbative collapse band; a perturbative regime governed by the signed first-order geometry term), while magnitudes and signs are family coefficients—the sharpest being the sign of the QA alignment term, which predicts on which family pruning improves retrieval-style capability. Two findings frame the road ahead: the mid-scale robustness peak reproduced in two families, whose mechanism our retired concentration predictor does not explain, and the collapse boundary $d_c^*(N_0, f)$, which the completed distillation-and-recovery grid will attack with the same pre-registered discipline. The deliverables—reduced-form laws in basic parameters and the method-selection map—aim to make “what should this agent scale down from, and how?” a computed answer rather than a folklore one.

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A Appendix

A.1 Measurement details

Probe construction, tokenization and truncation rules, estimation/measurement splits, and streaming estimators for $\bar{g}_c$, F_c quadratic forms, and binned spectra will be documented here, together with the full per-density loss tables for each model size.

A.2 Re-analysis details

Recipe-stratified anchor tables, held-out transfer-law errors, the trained/transfer split analysis, 1PL/2PL comparisons, and the item-level shift-versus-deletion likelihood tests.

A.3 Pre-registration

Timestamped commits of functional forms, claim hierarchy (T1/T2), discriminating experiments, and the falsifiable concentration prediction for 8B/14B/32B.